

Fruit for Thought: Effects of Background Conditions on Machine Learning Fruit Classification

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Introduction

- Fruit classification is important in agriculture, industry, and technology.
- A K-Nearest Neighbors (KNN) model was developed using the Fruit-360 dataset.
- The model achieved an almost perfect accuracy under controlled conditions.
- However, real-world images may differ, raising concerns about performance in real-world conditions.

Motivation And Hypothesis

- The model was trained using images with a uniform white background.
 - Real-world images often have more complex backgrounds.
 - This may limit the model's ability to perform well in different environments.

Hypothesis: Background conditions influence classification performance, and accuracy may decrease as background complexity increases.

Data Source

The training dataset was obtained from Kaggle, specifically the **Fruit 360 dataset**, which provides labeled fruit images under controlled conditions shown in the Experimental Design.

A separate testing dataset was collected manually under



Pre-processing

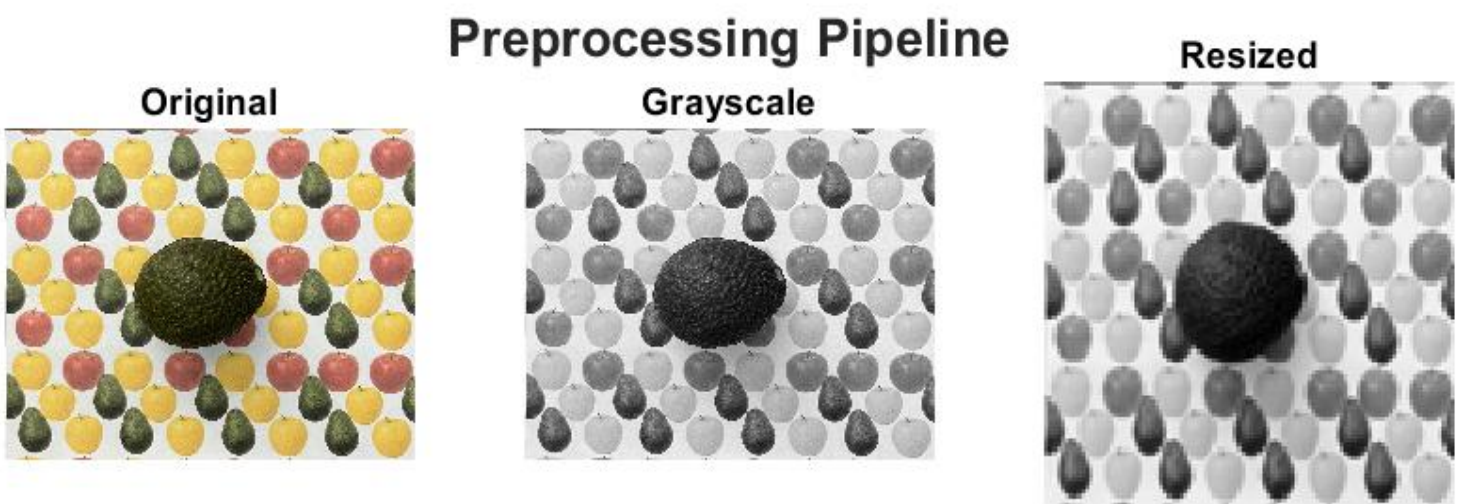
Approach 1: Grayscale

Images were preprocessed to ensure consistent input for classification.

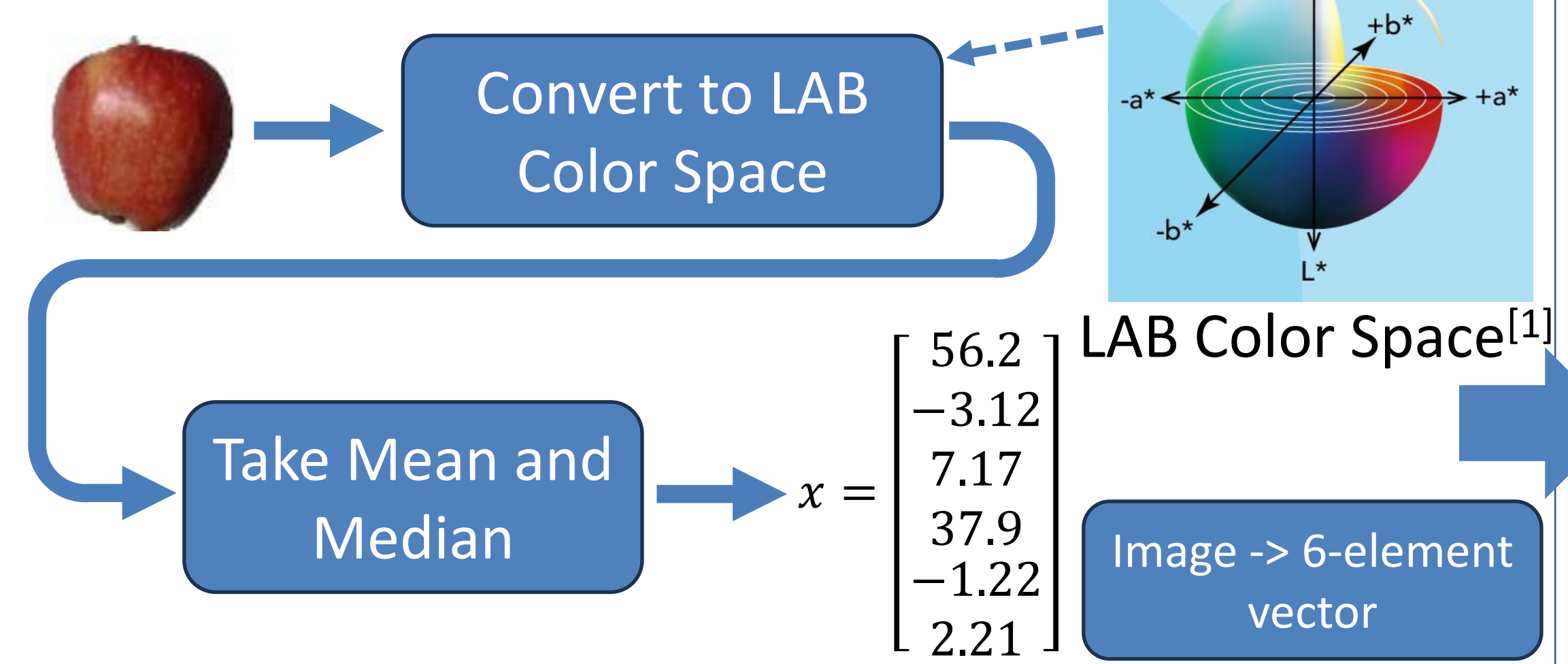
Steps include:

- Resizing to a fixed resolution
- Conversion to grayscale
- Vectorization into feature vectors

No cropping was applied during training and testing, so background information remains part of the input.



Approach 2: LAB Color Space



- 2 models used:
- KNN (15 neighbors)
 - Random Forest (1000 trees)

Cross-validation^[2]:

- 5-fold k-folding on training data (100% avg. accuracy, both models)
- Final models built with full training dataset

Results

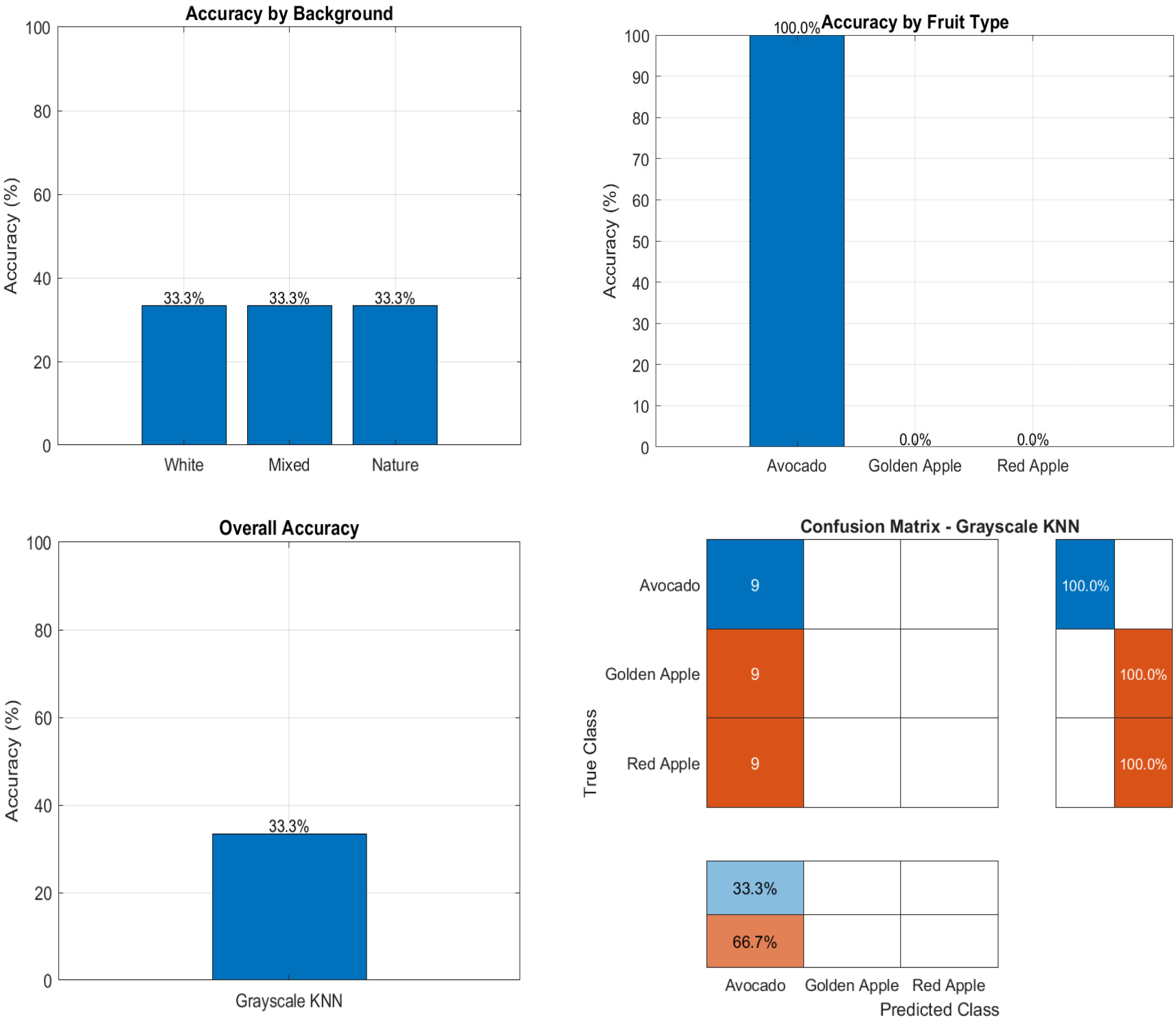
Approach 1 Results

Grayscale KNN Results:

- Accuracy $\approx$ 33.33% (near chance for 3 classes)
- Similar results after cropping
- Model did not generalize to new images

Interpretation:

- Strong dependence on dataset-specific features
- Sensitive to lighting and image condition



Approach 2 Results

89% accuracy on both models (24/27 correct)

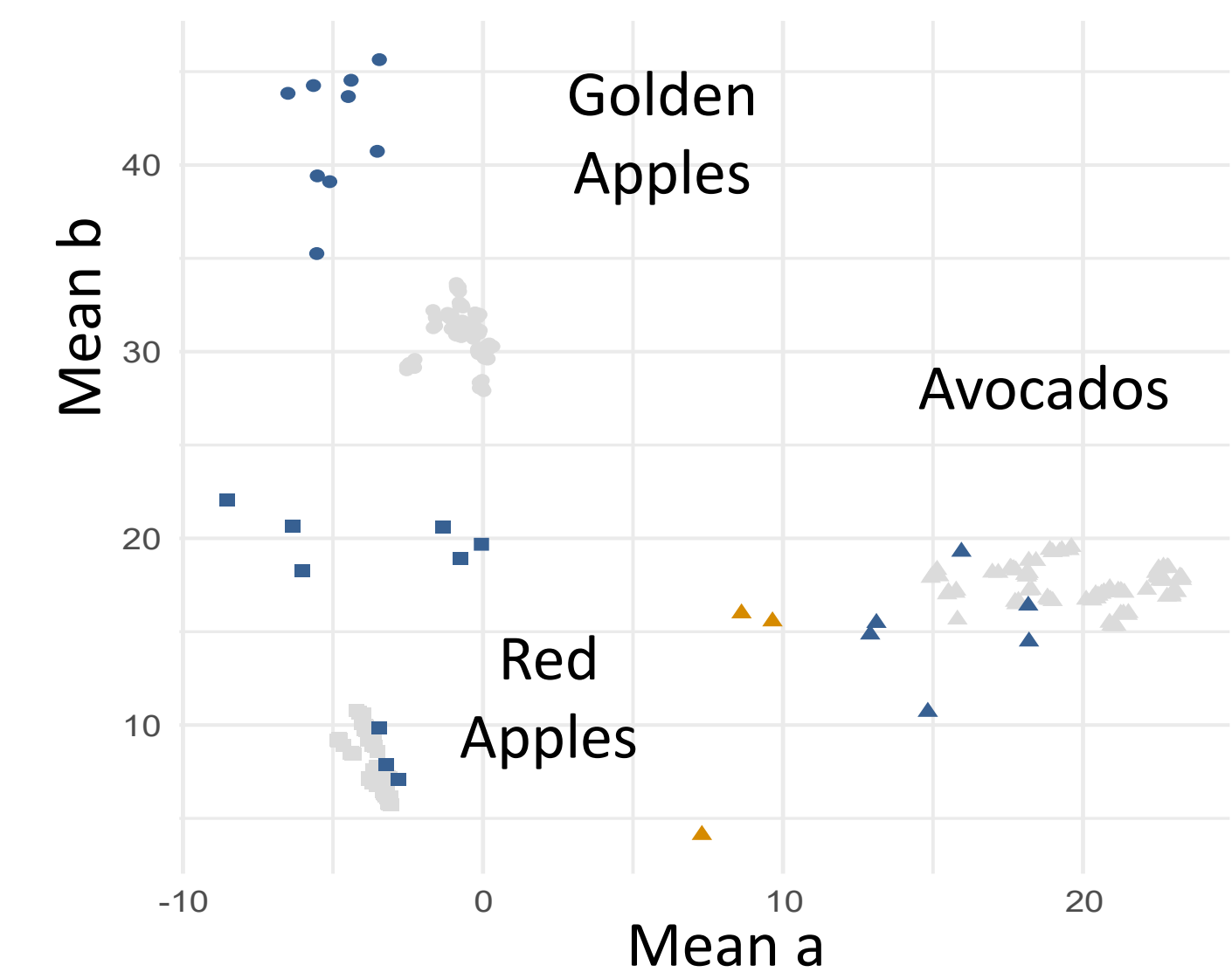
Models confused avocado and red apple for ...

Random Forest:
2x Avocado/Mixed
1x Red Apple/White

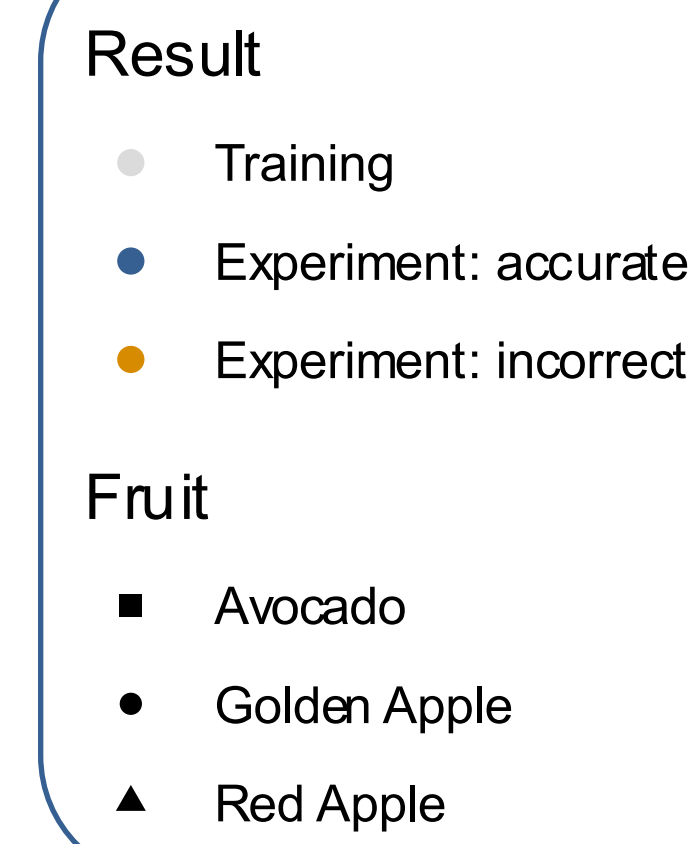
KNN:
2x Red Apple/Nature
1x Red Apple/White

Most misclassifications on complicated backgrounds!

KNN Misclassification (Feature Subspace)



Misclassification between clusters



Experimental Design



Fruit and Background

Fruit	Background
Golden Apple	White
Avocado	Mixed
Red Apple	Nature

Factors:

- Fruit type (3 levels): Avocado, Golden Apple, Red Apple
- Background (3 levels): White, Mixed, Nature

Each fruit type was evaluated across all background conditions, and each condition was replicated three times with randomized trial order.

Design structure:

For each trial:

- A different sample of each fruit was selected
- Each fruit type was evaluated across all three background conditions
- This results in a 3 x 3 combination (9 conditions per replicate)

Evaluation Approach:

- The model was trained on open-source images
- Performance was evaluated on independently collected test images
- Accuracy was analyzed across:
 - Fruit type
 - Background condition

Discussion & Conclusions

- Using LAB color space features (Approach 2) improved classification performance on experimental images.
- Under controlled camera and lighting conditions:
 - Approach 1: differences in background between the training and testing datasets likely has a strong effect on the model performance.
 - Approach 2: changing the background type (white, nature, mixed) in the testing dataset influences LAB-based features and contributes to misclassification between avocados

Future Direction

Future work will evaluate the model under more realistic background conditions and further assess its robustness.

References and Acknowledgement

- [1] <https://www.hunterlab.com/blog/what-is-cielab-color-space/>
[2] https://scikit-learn.org/stable/modules/cross_validation.html
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